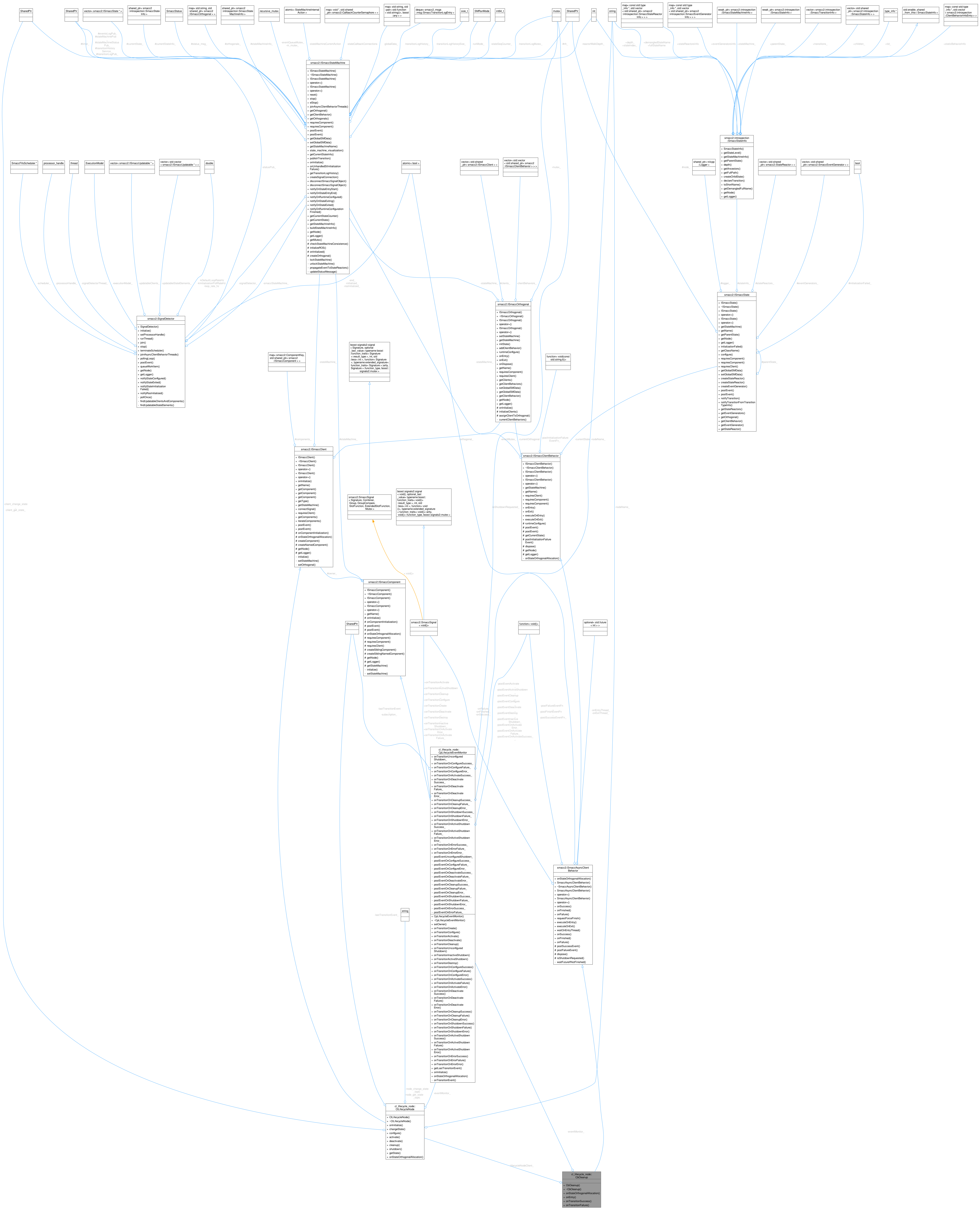
SharedPtr
SharedPtr
vector< smacc2::ISmaccState * >
shared_ptr< smacc2::introspection::SmaccStateInfo >
SmaccStatus
map< std::string, std::shared_ptr< smacc2::ISmaccOrthogonal > >
shared_ptr< smacc2::introspection::SmaccStateMachineInfo >
recursive_mutex
atomic< StateMachineInternalAction >
map< void *, std::shared_ptr< smacc2::CallbackCounterSemaphore > >
map< std::string, std::pair< std::function< std::string()>, boost::any > >
deque< smacc2_msgs::msg::SmaccTransitionLogEntry >
size_t
SMRunMode
int64_t
mutex
SharedPtr
int
string
map< const std::type_info *, std::vector< std::shared_ptr< smacc2::introspection::SmaccStateReactorInfo > > >
map< const std::type_info *, std::vector< std::shared_ptr< smacc2::introspection::SmaccEventGeneratorInfo > > >
weak_ptr< smacc2::introspection::SmaccStateMachineInfo >
weak_ptr< smacc2::introspection::SmaccStateInfo >
vector< smacc2::introspection::SmaccTransitionInfo >
vector< std::shared_ptr< smacc2::introspection::SmaccStateInfo > >
type_info *
std::enable_shared_from_this< SmaccStateInfo >
map< const std::type_info *, std::vector< smacc2::introspection::ClientBehaviorInfoEntry > >
#timer_
#eventsLogPub_ #stateMachinePub_ #stateMachineStatusPub_ #transitionHistoryService_ #transitionLogPub_
#currentState_
#currentStateInfo_
#status_msg_
#orthogonals_
#stateMachineInfo_
-eventQueueMutex_ -m_mutex_
-stateMachineCurrentAction
-stateCallbackConnections
-globalData_
-transitionLogHistory_
-transitionLogHistorySize_
-runMode_
-stateSeqCounter_
-transitionLogMutex_
#nh_
-reactorWalkDepth_
+depth_ +stateIndex_
+demangledStateName +fullStateName
+stateReactorsInfo
+eventGeneratorsInfo
+stateMachine_
+parentState_
+transitions_
+children_
+tid_
+staticBehaviorInfo
smacc2::ISmaccStateMachine
+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ ISmaccStateMachine()
+ operator=()
+ ISmaccStateMachine()
+ operator=()
+ reset()
+ stop()
+ eStop()
+ joinAsyncClientBehaviorThreads()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ onUnhandledStInitializationFailure()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfigurationFinished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()
smacc2::introspection::SmaccStateInfo
+ SmaccStateInfo()
+ getStateLevel()
+ getStateMachineInfo()
+ getParentState()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()
-statusPub_
-mutex_
#node_
SmaccFifoScheduler *
processor_handle
thread
ExecutionModel
vector< smacc2::ISmaccUpdatable * >
vector< std::vector< smacc2::ISmaccUpdatable * > >
double
atomic< bool >
vector< std::shared_ptr< smacc2::ISmaccClient > >
vector< std::vector< std::shared_ptr< smacc2::ISmaccClientBehavior > > >
shared_ptr< rclcpp::Logger >
vector< std::shared_ptr< smacc2::StateReactor > >
vector< std::shared_ptr< smacc2::SmaccEventGenerator > >
bool
-scheduler_
-processorHandle_
-signalDetectorThread_
-executionModel_
-updatableClients_
-updatableStateElements_
-kDefaultLoopRateHz -kInitializationPollRateHz -loop_rate_hz
-signalDetector_
-smaccStateMachine_
-end_ -initialized_ -rosInitialized_
-stateMachine_
#clients_
-clientBehaviors_
#logger_
#stateInfo_
#stateReactors_
#eventGenerators_
#initializationFailed_
smacc2::SignalDetector
+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ joinAsyncClientBehaviorThreads()
+ pollingLoop()
+ postEvent()
+ queueWorkItem()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyStateInitializationFailed()
+ notifyRosInitialized()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()
smacc2::ISmaccOrthogonal
+ ISmaccOrthogonal()
+ ~ISmaccOrthogonal()
+ ISmaccOrthogonal()
+ operator=()
+ ISmaccOrthogonal()
+ operator=()
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# onInitialize()
# initializeClients()
# assignClientToOrthogonal()
- currentClientBehaviors()
smacc2::ISmaccState
+ ISmaccState()
+ ~ISmaccState()
+ ISmaccState()
+ operator=()
+ ISmaccState()
+ operator=()
+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ initializationFailed()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransition()
+ notifyTransitionFromTransitionTypeInfo()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()
#parentState_
function< void(const std::string &)>
map< smacc2::ComponentKey, std::shared_ptr< smacc2::ISmaccComponent > >
boost::signals2::signal< Signature, optional_last_value< typename boost::function_traits< Signature >::result_type >, int, std::less< int >, function< Signature >, typename extended_signature< function_traits< Signature >::arity, Signature >::function_type, boost::signals2::mutex >
-stateMachine_
-stateMachine_
#components_
#stateMachine_
-orthogonal_
-eventMutex_
-currentOrthogonal
-postInitializationFailureEventFn_
-currentState
-nodeName_
smacc2::ISmaccClient
+ ISmaccClient()
+ ~ISmaccClient()
+ ISmaccClient()
+ operator=()
+ ISmaccClient()
+ operator=()
+ onInitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvent()
# onComponentInitialization()
# onStateOrthogonalAllocation()
# createComponent()
# createNamedComponent()
# getNode()
# getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()
smacc2::ISmaccClientBehavior
+ ISmaccClientBehavior()
+ ~ISmaccClientBehavior()
+ ISmaccClientBehavior()
+ operator=()
+ ISmaccClientBehavior()
+ operator=()
+ getStateMachine()
+ getName()
+ requiresClient()
+ requiresComponent()
+ requiresComponent()
+ onEntry()
+ onExit()
+ executeOnEntry()
+ executeOnExit()
# runtimeConfigure()
# postEvent()
# postEvent()
# getCurrentState()
# postInitializationFailureEvent()
# dispose()
# getNode()
# getLogger()
- onStateOrthogonalAllocation()
-client_change_state_
-client_get_state_
-isShutdownRequested_
-nodeName_
smacc2::SmaccSignal< Signature, Combiner, Group, GroupCompare, SlotFunction, ExtendedSlotFunction, Mutex >
boost::signals2::signal< void(), optional_last_value< typename boost::function_traits< void()>::result_type >, int, std::less< int >, function< void()>, typename extended_signature< function_traits< void()>::arity, void()>::function_type, boost::signals2::mutex >
#owner_
< void()>
smacc2::ISmaccComponent
+ ISmaccComponent()
+ ~ISmaccComponent()
+ ISmaccComponent()
+ operator=()
+ ISmaccComponent()
+ operator=()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()
SharedPtr
smacc2::SmaccSignal< void()>
function< void()>
optional< std::future< int > >
+onTransitionActivate
+onTransitionActiveShutdown
+onTransitionCleanup
+onTransitionConfigure
+onTransitionCreate
+onTransitionDeactivate
+onTransitionDestroy
+onTransitionInactiveShutdown_
+onTransitionOnActivateError_
+onTransitionOnActivateFailure_
...
-lastTransitionEvent
-subscription_
-onFailure_ -onFinished_ -onSuccess_
-postEventActivate
-postEventActiveShutdown
-postEventCleanup
-postEventConfigure
-postEventDeactivate
-postEventDestroy
-postEventInactiveShutdown_
-postEventOnActivateError_
-postEventOnActivateFailure_
-postEventOnActivateSuccess_
...
-postFailureEventFn
-postFinishEventFn
...
-postSuccessEventFn_
-onEntryThread_ -onExitThread_
cl_lifecycle_node::CpLifecycleEventMonitor
+ onTransitionUnconfiguredShutdown_
+ onTransitionOnConfigureSuccess_
+ onTransitionOnConfigureFailure_
+ onTransitionOnConfigureError_
+ onTransitionOnActivateSuccess_
+ onTransitionOnDeactivateSuccess_
+ onTransitionOnDeactivateFailure_
+ onTransitionOnDeactivateError_
+ onTransitionOnCleanupSuccess_
+ onTransitionOnCleanupFailure_
+ onTransitionOnCleanupError_
+ onTransitionOnShutdownSuccess_
+ onTransitionOnShutdownFailure_
+ onTransitionOnShutdownError_
+ onTransitionOnActiveShutdownSuccess_
+ onTransitionOnActiveShutdownFailure_
+ onTransitionOnActiveShutdownError_
+ onTransitionOnErrorSuccess_
+ onTransitionOnErrorFailure_
+ onTransitionOnErrorError_
- postEventUnconfiguredShutdown_
- postEventOnConfigureSuccess_
- postEventOnConfigureFailure_
- postEventOnConfigureError_
- postEventOnDeactivateSuccess_
- postEventOnDeactivateFailure_
- postEventOnDeactivateError_
- postEventOnCleanupSuccess_
- postEventOnCleanupFailure_
- postEventOnCleanupError_
- postEventOnShutdownSuccess_
- postEventOnShutdownFailure_
- postEventOnShutdownError_
- postEventOnErrorSuccess_
- postEventOnErrorFailure_
+ CpLifecycleEventMonitor()
+ ~CpLifecycleEventMonitor()
+ setOwner()
+ onTransitionCreate()
+ onTransitionConfigure()
+ onTransitionActivate()
+ onTransitionDeactivate()
+ onTransitionCleanup()
+ onTransitionUnconfiguredShutdown()
+ onTransitionInactiveShutdown()
+ onTransitionActiveShutdown()
+ onTransitionDestroy()
+ onTransitionOnConfigureSuccess()
+ onTransitionOnConfigureFailure()
+ onTransitionOnConfigureError()
+ onTransitionOnActivateSuccess()
+ onTransitionOnActivateFailure()
+ onTransitionOnActivateError()
+ onTransitionOnDeactivateSuccess()
+ onTransitionOnDeactivateFailure()
+ onTransitionOnDeactivateError()
+ onTransitionOnCleanupSuccess()
+ onTransitionOnCleanupFailure()
+ onTransitionOnCleanupError()
+ onTransitionOnShutdownSuccess()
+ onTransitionOnShutdownFailure()
+ onTransitionOnShutdownError()
+ onTransitionOnActiveShutdownSuccess()
+ onTransitionOnActiveShutdownFailure()
+ onTransitionOnActiveShutdownError()
+ onTransitionOnErrorSuccess()
+ onTransitionOnErrorFailure()
+ onTransitionOnErrorError()
+ getLastTransitionEvent()
+ onInitialize()
+ onStateOrthogonalAllocation()
- onTransitionEvent()
smacc2::SmaccAsyncClientBehavior
+ onStateOrthogonalAllocation()
+ SmaccAsyncClientBehavior()
+ ~SmaccAsyncClientBehavior()
+ SmaccAsyncClientBehavior()
+ operator=()
+ SmaccAsyncClientBehavior()
+ operator=()
+ onSuccess()
+ onFinished()
+ onFailure()
+ requestForceFinish()
+ executeOnEntry()
+ executeOnExit()
+ waitOnEntryThread()
+ onSuccess()
+ onFinished()
+ onFailure()
# postSuccessEvent()
# postFailureEvent()
# dispose()
# isShutdownRequested()
- waitFutureIfNotFinished()
string
-lastTransitionEvent_
-node_change_state_topic
-node_get_state_topic
-eventMonitor_
cl_lifecycle_node::ClLifecycleNode
+ ClLifecycleNode()
+ ~ClLifecycleNode()
+ onInitialize()
+ changeState()
+ configure()
+ activate()
+ deactivate()
+ cleanup()
+ shutdown()
+ getState()
+ onStateOrthogonalAllocation()
-eventMonitor_
-lifecycleNodeClient_
cl_lifecycle_node::CbCleanup
+ CbCleanup()
+ ~CbCleanup()
+ onStateOrthogonalAllocation()
+ onEntry()
+ onTransitionSuccess()
+ onTransitionFailure()